

TED(15)-3213
(Revision -2015)

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THIRD SEMESTER DIPLOMA EXAMINATIONS IN INSTRUMENTATION
ENGINEERING- MODEL QUESTION PAPER

ELECTRICAL AND ELECTRONIC INSTRUMENTS

[Time: 3 hours]

(Maximum marks :100)

PART -A

(Maximum marks :10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. State the principle of moving coil instruments.
2. Mention the use of copper bands on the limbs of voltage magnet in an induction type energymeter.
3. List any four parts of Cathode ray tube.
4. Define deflection sensitivity.
5. State principle employed in a Ramp type Digital Voltmeter.

PART-B

(Maximum marks :30)

II. Answer any **five** of the following questions. Each question carries six marks.

- a. Distinguish between Moving iron repulsion type instrument and Moving iron attraction type instrument.
- b. Justify the use of 'D' A rsonval movement in the analog multimeters.
- c. Illustrate the impedance measurement using Maxwell's Bridge.
- d. Comment on the functions of basic controls provided at the front panel of a Cathode ray oscilloscope.
- e. Explain a sampling oscilloscope with the help of a neat block diagram.
- f. Explain X-Y recorder.
- g. Compare the application of a strip chart recoder and a circular chart recorder.

PART-C

(Maximum marks :60)

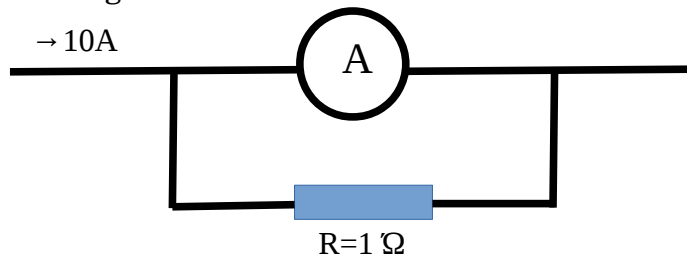
Answer one full question from each unit. Each full question carries 15 marks.

MODULE - I

III.

- a. Comment on the construction of the coils of the moving iron type ammeter and the voltmeter. (7marks)

- b. Modify the following scheme to extend the range of the ammeter to 100A.
Current rating of the ammeter coil is 100mA. (8 marks)



OR

IV.

- Draw the schematic diagram of the arrangement where a multimeter is intended to show the resistance of a 10 ohm resistor. (7 marks)
- Illustrate the construction of a moving iron repulsion type voltmeter. (8 marks)

MODULE - II

V.

- Draw the circuit diagram of connecting a 300V, 5A Wattmeter in a 230V, 50 Hz. single phase motor circuit and predict the behaviour of the meter when the motor is switched on. (7 marks)
- Demonstrate the working of a Wien bridge when it is used for the measurement of capacitance of a cable. (8marks)

OR

VI.

- Explain the working of Hay's bridge with the help of a neat sketch. (7 marks)
- Illustrate the construction of potential magnet of a single phase induction type energy meter. (8 marks)

MODULE-III

VII.

- Explain electrostatic deflection used in the Cathode ray tube. (7 marks)
- Demonstrate the method of measuring voltage and frequency of 10V, 50Hz. AC supply using a CRO. (8 marks)

OR

VIII.

- Explain any one type of CRO probe. (7 marks)
- Compare Dual trace and dual beam Cathode ray oscilloscopes. (8 marks)

MODULE-IV

IX.

- a. Explain the working of Galvanometric type recorder. (7 marks)
- b. Explain a strip chart recorder with block diagram . (8 marks)

OR

X.

- a. Explain the digital multimeter with block diagram. (7 marks)
- b. Make comments on the analog to digital conversion technique used in a Ramp type Digital voltmeter. (8 marks)